

Brief Report

What include

- Docker Segmentation Process and Implementation
- Applicability Compliance Lists
- Time Planning Table

1. Docker Segmentation Process and Implementation

Containers / log-server

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log-server

 4135b9a2f4f4  alpine:3.23

Logs

Inspect

Bind mounts

Exec

Files

Stats

(1/4) Installing ncurses-terminfo-base (6.5_p20251123-r0)

(2/4) Installing libncursesw (6.5_p20251123-r0)

(3/4) Installing readline (8.3.1-r0)

(4/4) Installing socat (1.8.0.3-r1)

Executing busybox-1.37.0-r30.trigger

OK: 9464 KiB in 20 packages

2026/06/06 17:24:53 socat[1] W address is opened in read-write mode but only supports read-only

2026/06/06 17:24:53 socat[1] W address is opened in read-write mode but only supports write-only

test log event

> 2026/06/06 17:38:12.000474253 length=15 from=0 to=14

test log event

(Log Forwarding Testing : Simplex Mode)

<

 s2

D:\Workspaces\Test\S2

View configurations







 log-server

alpine:3.23







 db

postgres:18.4







 kms

alpine:3.23







 pos

alpine:3.23







 app-server

alpine:3.23







 internal-fw

s2-internal-fw







 perimeter-fw

s2-perimeter-fw







stderr: warning: enabling 'strict' segmentation for 'core' connections

initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.

waiting for server to start....2026-06-06 17:24:53.813 UTC [54] LOG: starting PostgreSQL 18.4 (Debian 18.4-1.pgdb13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit

2026-06-06 17:24:53.820 UTC [54] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"

2026-06-06 17:24:53.868 UTC [60] LOG: database system was shut down at 2026-06-06 17:24:53 UTC

2026-06-06 17:24:53.879 UTC [54] LOG: database system is ready to accept connections

done

server started

CREATE DATABASE

/usr/local/bin/docker-entrypoint.sh: ignoring /docker-entrypoint-initdb.d/*

waiting for server to shut down....2026-06-06 17:24:54.235 UTC [54] LOG: received fast shutdown request

2026-06-06 17:24:54.242 UTC [54] LOG: aborting any active transactions

2026-06-06 17:24:54.246 UTC [54] LOG: background worker "logical replication launcher" (PID 63) exited with exit code 1

2026-06-06 17:24:54.246 UTC [58] LOG: shutting down

2026-06-06 17:24:54.254 UTC [58] LOG: checkpoint starting: shutdown immediate

2026-06-06 17:24:54.412 UTC [58] LOG: checkpoint complete: wrote 943 buffers (5.8%), wrote 3 SLRU buffers; 0 WAL file(s) added, 0 removed, 0 recycled; write=0.840 s, sync=0.093 s, total=0.166 s; sync files=303, longest=0.029 s, average=0.001 s; distance=4352 kB, estimate=4352 kB; lsn=0/1B9FBA8, redo lsn=0/1B9FBA8

2026-06-06 17:24:54.440 UTC [54] LOG: database system is shut down

done

server stopped

PostgreSQL init process complete; ready for start up.

2026-06-06 17:24:54.586 UTC [1] LOG: starting PostgreSQL 18.4 (Debian 18.4-1.pgdb13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit

2026-06-06 17:24:54.587 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432

2026-06-06 17:24:54.587 UTC [1] LOG: listening on IPv6 address ":::", port 5432

2026-06-06 17:24:54.602 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"

2026-06-06 17:24:54.622 UTC [76] LOG: database system was shut down at 2026-06-06 17:24:54 UTC

2026-06-06 17:24:54.633 UTC [1] LOG: database system is ready to accept connections

2026-06-06 17:29:54.705 UTC [74] LOG: checkpoint starting: time

2026-06-06 17:29:59.121 UTC [74] LOG: checkpoint complete: wrote 43 buffers (0.3%), wrote 3 SLRU buffers; 0 WAL file(s) added, 0 removed, 0 recycled; write=4.341 s, sync=0.026 s, total=4.417 s; sync files=12, longest=0.016 s, average=0.003 s; distance=318 kB, estimate=318 kB; lsn=0/1BEF630, redo lsn=0/1BEF508

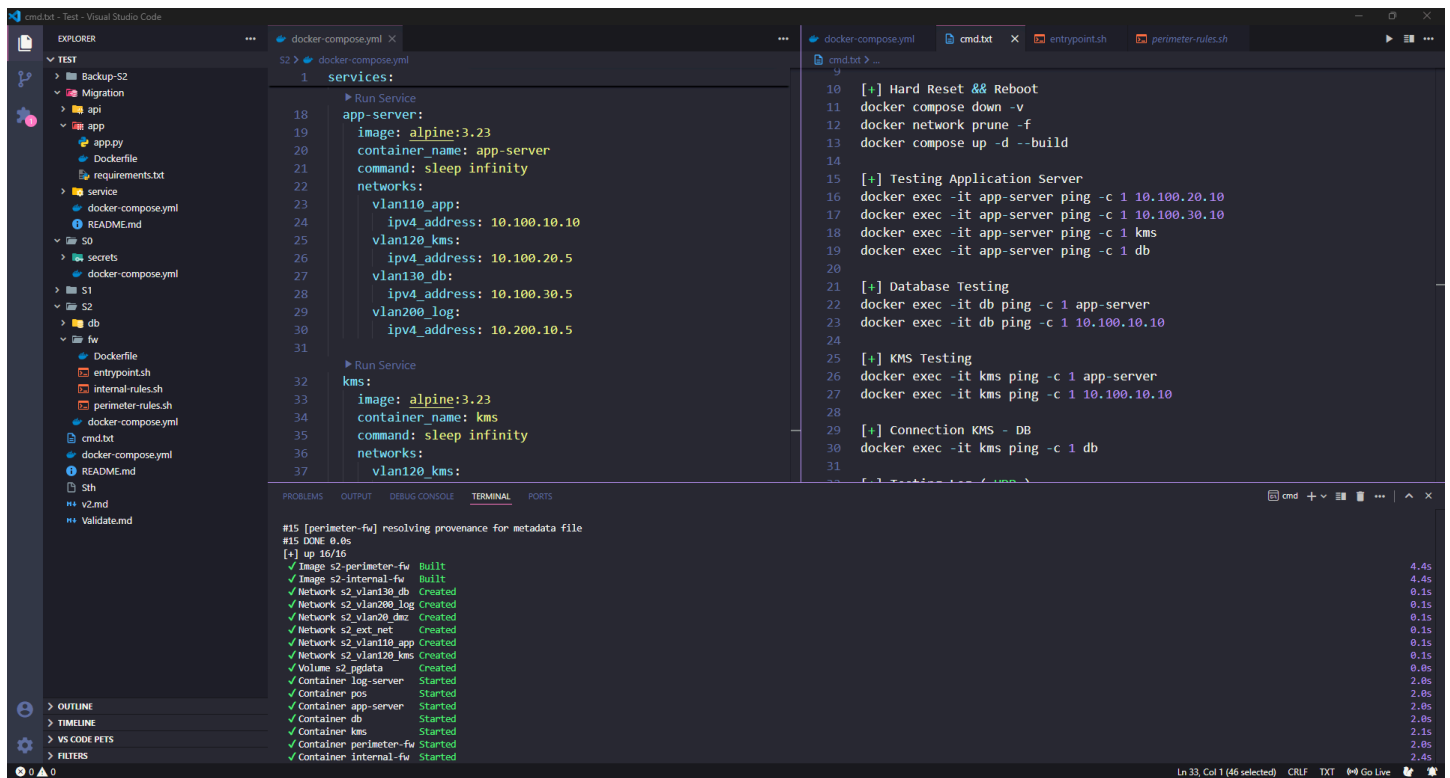
log-server

test log event

> 2026/06/06 17:38:12.000474253 length=15 from=0 to=14

test log event

(Container's Components Without Departments)



(Code and Testing Validation Sample Commands)

```
C:\Users\t0fu>docker exec -it app-server ping -c 1 kms
PING kms (10.100.20.10): 56 data bytes
64 bytes from 10.100.20.10: seq=0 ttl=64 time=0.244 ms
```

```
--- kms ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.244/0.244/0.244 ms
```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug app-server](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

```
C:\Users\t0fu>docker exec -it db ping -c 1 app-server
OCI runtime exec failed: exec failed: unable to start container process: exec: "ping": executable file not found in $PATH
```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug db](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

```
C:\Users\t0fu>docker exec -it kms ping -c 1 app-server
PING app-server (10.100.20.5): 56 data bytes
64 bytes from 10.100.20.5: seq=0 ttl=64 time=0.083 ms
```

```
--- app-server ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.083/0.083/0.083 ms
```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug kms](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

```
C:\Users\t0fu>docker exec -it kms ping -c 1 db
PING db (10.200.10.7): 56 data bytes
64 bytes from 10.200.10.7: seq=0 ttl=64 time=0.326 ms
```

```
--- db ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.326/0.326/0.326 ms
```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug kms](#)
Learn more at <https://docs.docker.com/go/debug-cli/>

(Validation on CDE)

2. Applicability Compliance Lists

Applicability Compliance		Summary of Applicable Requirements		Summary of Applicable Requirements	

As for right now I have established the baseline of what I’m choosing, apologized for not organizing it better before submission but it’s to have the professor review the draft first.

Note : Some requirements might be add or remove in the future 🙏

3. Time Planning Table : [Link](#)

Project Management Plan

PROJECT TITLE: Network Segmentation Effectiveness for PCI DSS Scope

STUDENT: Mr. Thy Dayuth

ADVISOR: Mr. Meas Sotheareth

COMPANY NAME: ACLEDA Bank Plc.

PROJECT SUPERVISOR: Mr. Phok Visal

WBS NUMBER	TASK TITLE	START DATE	COMPLETED DATE	DURATION	TASK STATUS	PHASE TWO				PHASE THREE				PHASE FOUR				PHASE FIVE				PHASE SIX				
						WEEK 6		WEEK 7		WEEK 8		WEEK 9		WEEK 10		WEEK 11		WEEK 12								
						M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M
A	Category A : Core Network & Infrastructure																									
A01	Configure Switch 1 Dept VLANs	25-May-26	27-May-26	2	COMPLETED																					
A02	Configure Switch 2 CDE Segmentation	27-Dec-99	29-Dec-99	2	COMPLETED																					
A03	Configure Inbound/Outbound Firewall Rules	29-May-26	31-May-26	2	COMPLETED																					
A04	Provision Perimeter FW (VLAN 10/20)	31-May-26	01-Jun-26	1	COMPLETED																					
A05	Provision Internal FW (CDE & Logs)	01-Jun-26	03-Jun-26	2	COMPLETED																					
B	Category B : Security Controls & Firewall Rules																									
B01	Implement Department Isolation ACLs	03-Jun-26	05-Jun-26	2	COMPLETED																					
B02	Configure Switch 1 Jumphost Routing			0	IN PROGRESS																					
B03	Establish Retailer-to-DMZ TLS 1.2 Tunnel			0	IN PROGRESS																					
B04	Restrict App Server KMS/DB Traffic	05-Jun-26	06-Jun-26	1	COMPLETED																					
B05	Enforce Deny-by-Default (FW & Switch 1)	06-Jun-26	09-Jun-26	3	COMPLETED																					
C	Category C : Systems & Identity Access Management (IAM)																									
C01	Deploy Jumphost (Static IP)			0	NOT STARTED																					
C02	Harden Jumphost Server			0	NOT STARTED																					
C03	Implement Jumphost MFA			0	NOT STARTED																					
C04	Deploy Logging Server Pipeline	04-Jun-26	07-Jun-26	3	IN PROGRESS																					
D	Category D : Database & Application Development																									
D01	Provision PostgreSQL & DB Roles			0	IN PROGRESS																					
D02	Initialize DB Schema			0	IN PROGRESS																					
D03	Integrate Key Management System (KMS)			0	IN PROGRESS																					
D04	Develop App Server Inbound Handling			0	IN PROGRESS																					
D05	Integrate App Server KMS API			0	IN PROGRESS																					
D06	Develop App Server DB Persistence Layer			0	IN PROGRESS																					
D07	Develop POS Payload Generation Script			0	IN PROGRESS																					
E	Category E : Integration, Testing & End-to-End Validation																									
E01	Conduct Network Isolation Nmap Scan			0	NOT STARTED																					
E02	Verify MFA Integration			0	NOT STARTED																					
E03	Verify CDE & KMS/DB Access Controls			0	NOT STARTED																					
E04	Simulate End-to-End POS Transaction			0	NOT STARTED																					
E05	Validate Security Audit Logs			0	NOT STARTED																					
F	Category F : Documentation & Reporting (Final Phase)																									
F01	Finalize Project Report			7 days	NOT STARTED																					
					NOT STARTED																					

As for now, I have completed some of the core which is the structure both starting on the CDE code as well, I’ll keep it up to date as much as possible.

Note : Some are planning, Some finished for a phase but still need monitoring in case there’s modification